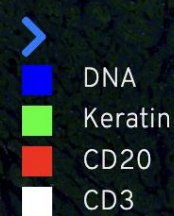


Intro to spatial 'omics



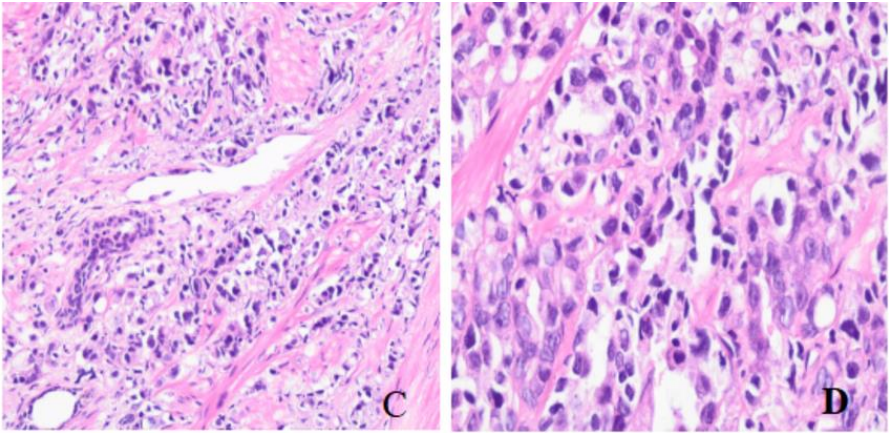
Day 1

1. Histology
2. Spatial proteomics
3. Live demo

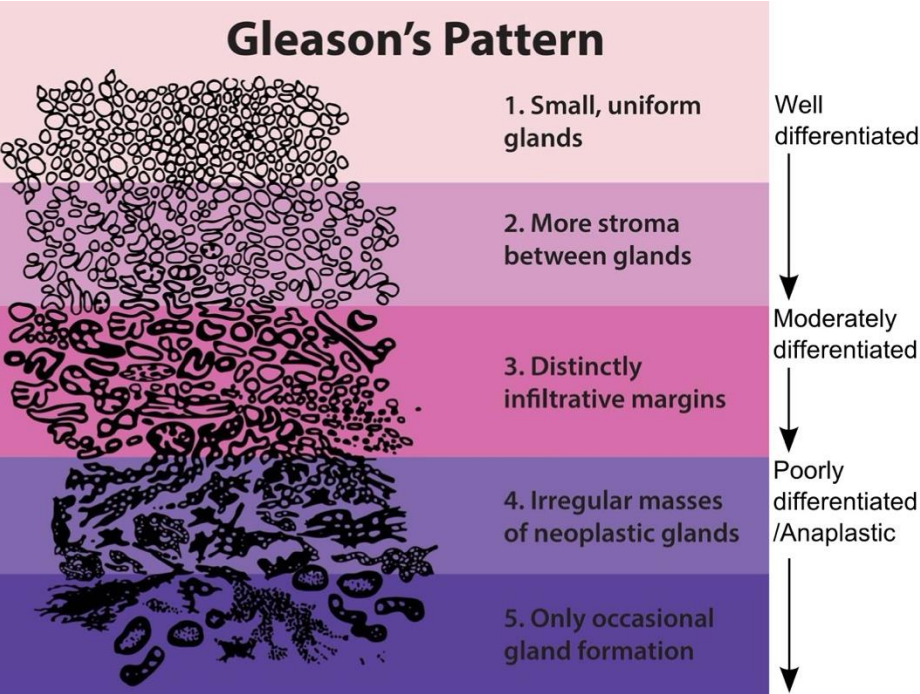
Day 2

1. Spatial transcriptomics
2. Live demo

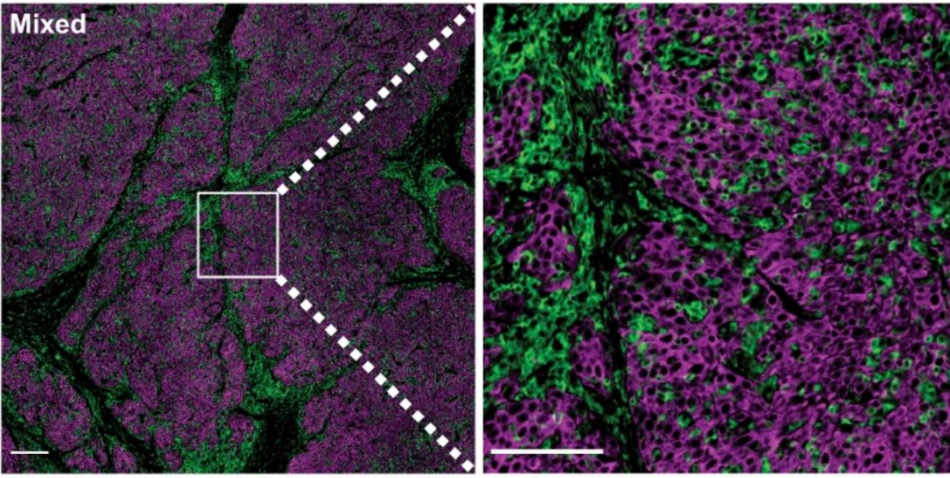
Review of last time



Epstein (2018) *Modern Pathology*

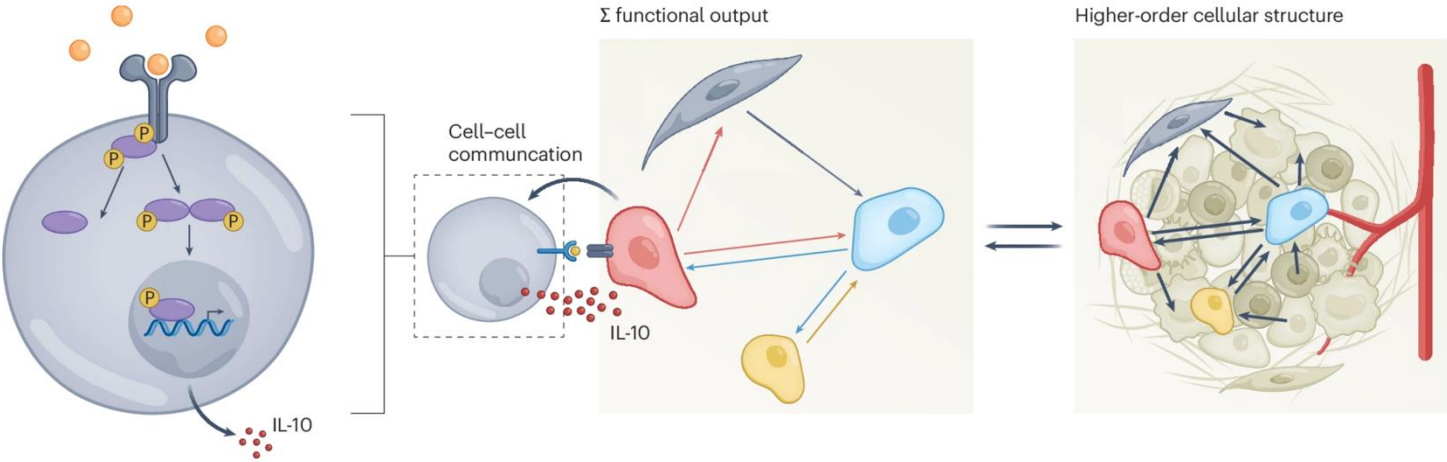


Morphology & Grade. ICD-O-3 Morphology Codes. *National Institutes of Health.*



Blise et al. (2022) *Nature Methods*

From: [Highly multiplexed imaging in the omics era: understanding tissue structures in health and disease](#)

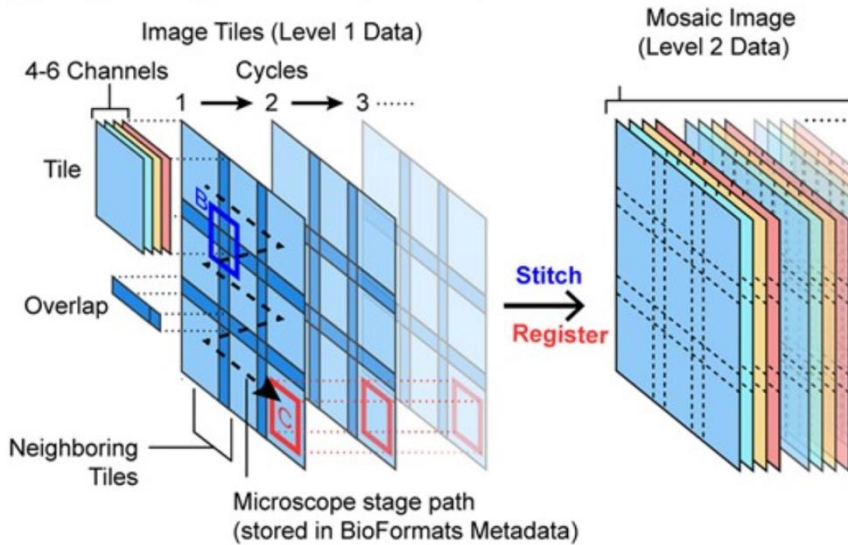


Bodenmiller (2024) *Nature Methods*

Review of last time: Computational processing

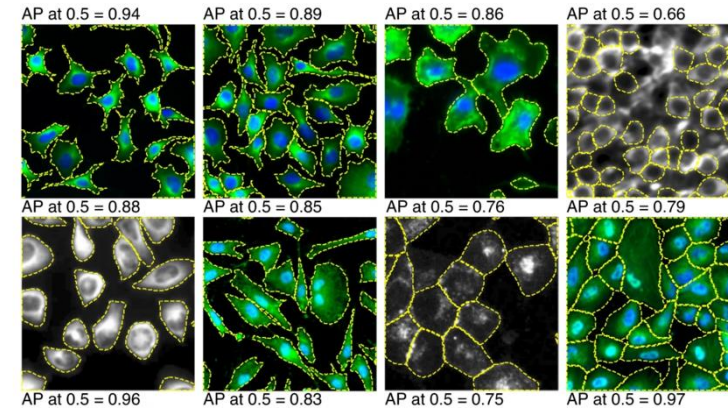
1

(a) Steps in imaging and data processing



2

From: [Cellpose: a generalist algorithm for cellular segmentation](#)



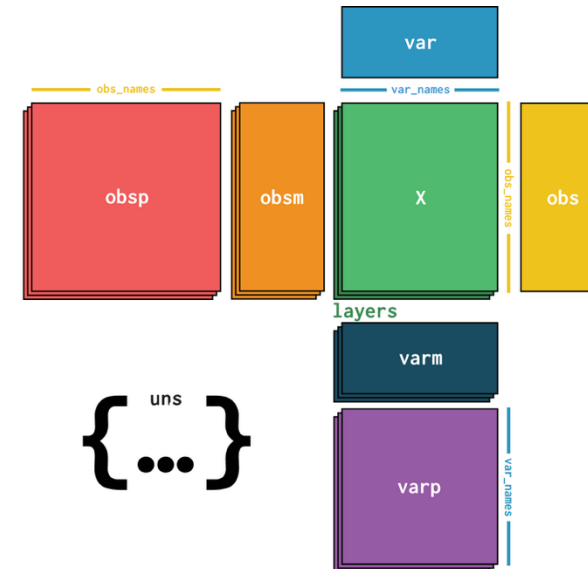
Stringer et al. 2020 *Nature Methods*

Muhlich et al. 2022 *Bioinformatics*

3

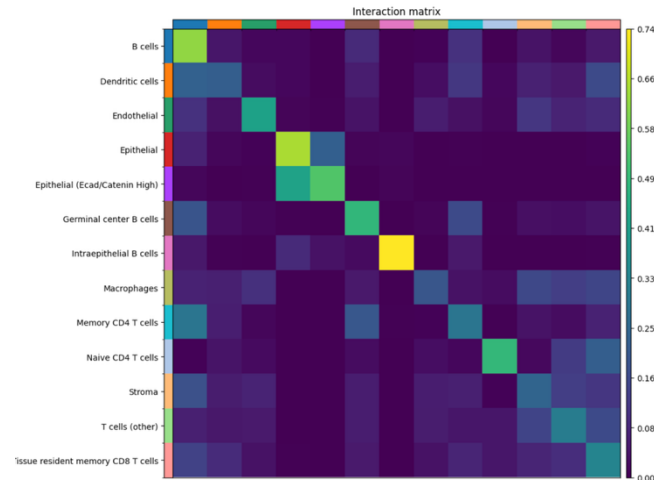
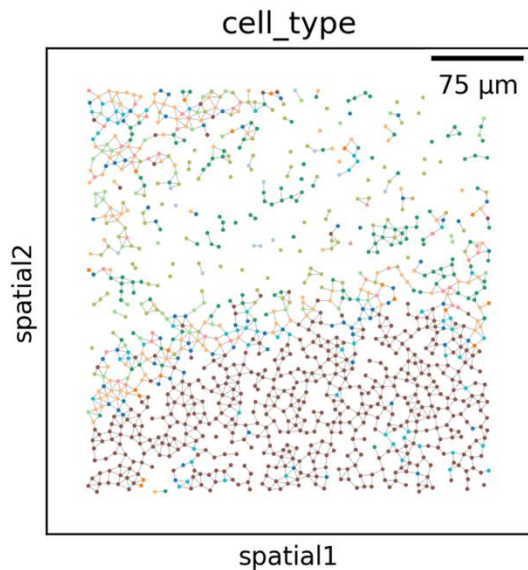
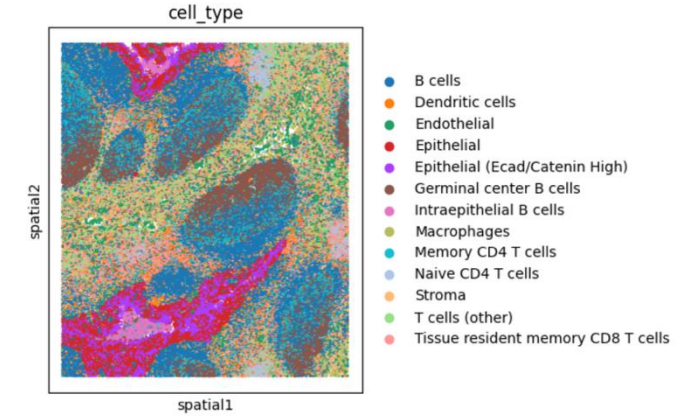
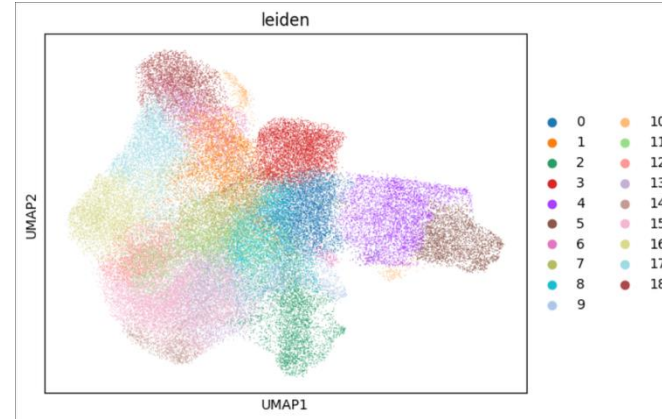
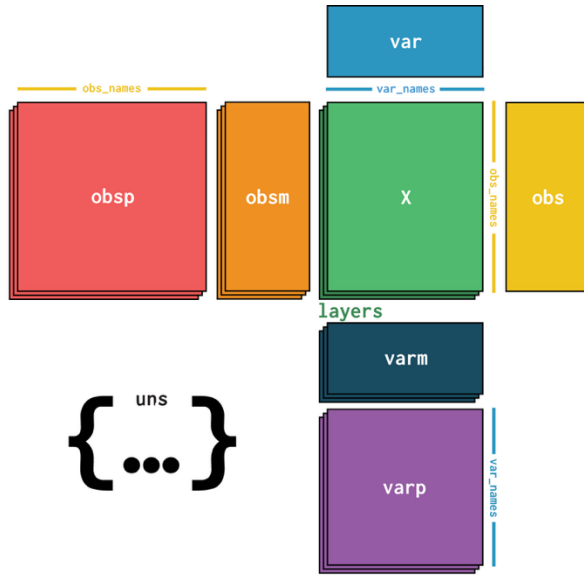
CellID	Hoechst1	anti_NaKATP	anti_CD3	anti_CD45RO	Ki67_488	Keratin_570
1	13673.167	3979.4167	914.08333	3444.583333	2842.3333	389.66667
2	13990.579	4874.2105	772.05263	2675.736842	2606.7895	1272.9474
3	10669.538	12600.231	938.46154	1149.846154	4988.9231	2073.3846
4	12799.917	3279.75	4803.5833	2860.75	2320.0833	311.08333
5	16906.704	5358.9259	16498	9213.333333	2852.3333	328.25926
6	18506.385	3474.359	6746.359	3828.102564	2702.2051	305.69231
7	8441	4879.9333	2175.2	1753.333333	2992.2	274.2
8	13036.8	3349.1	1079.9	1629.1	2490.8	355.85
9	13605.042	2826	1138.125	2688.041667	2388.9583	357.91667
10	17670.364	3020.0909	912.68182	1321	2577.5909	500.59091
11	13516.636	3582.4545	935.81818	1285.454545	2811.8182	368.45455
12	11877.644	3311.6889	997.68889	2274.111111	3334.6889	397.15556

4

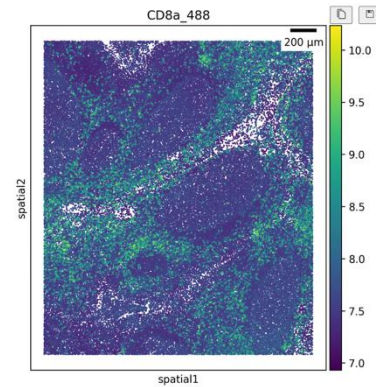
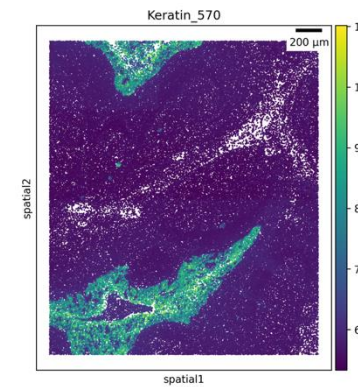


<https://anndata.readthedocs.io/en/stable/>

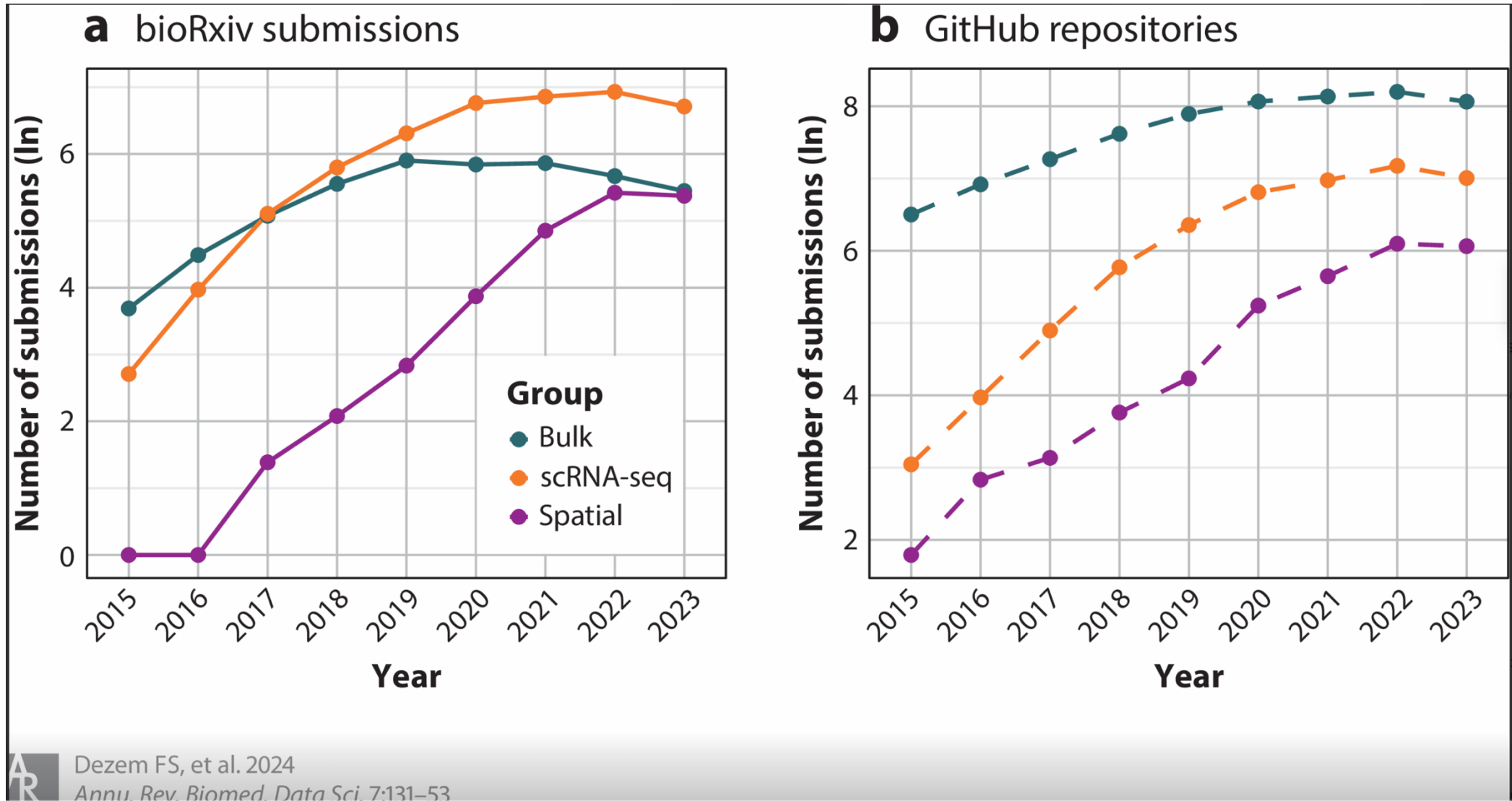
Review of last time: Downstream analysis



	I	pval_norm
Keratin_570	0.881100	0.0
Ecad_488	0.802425	0.0
CD20_488	0.652074	0.0
Ki67_488	0.635294	0.0
anti_NaKATPase	0.633640	0.0
aSMA_660	0.607259	0.0
Catenin_647	0.604994	0.0
CD4_570	0.578281	0.0
CD45_647	0.573716	0.0
anti_CD3	0.567418	0.0
PD1_647	0.552441	0.0
Vimentin_555	0.546819	0.0
CD31_647	0.536576	0.0
CD8a_488	0.531511	0.0
CD11c_555	0.515563	0.0
CD163_488	0.502560	0.0
anti_CD45RO	0.498790	0.0
CD68_488	0.388706	0.0



Spatial transcriptomics is becoming increasingly popular

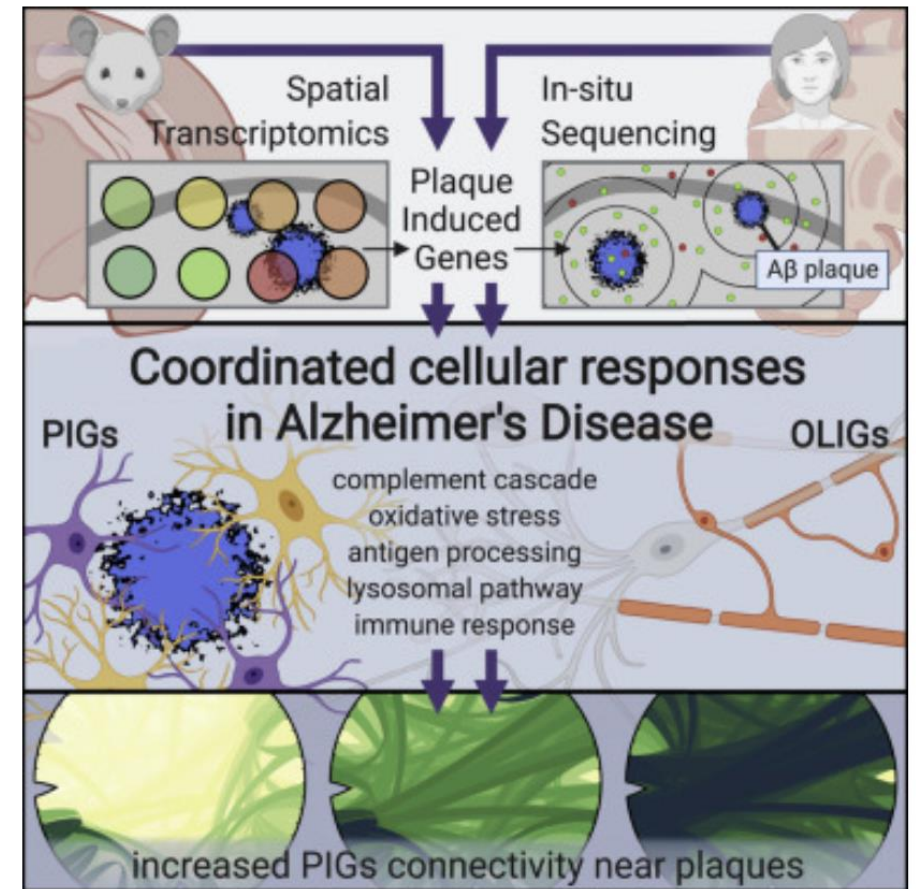


Why spatial transcriptomics?

- Spatial proteomics can tell us a lot about cell states and function, but is very limited in panel size
- scRNA-seq lacks spatial information and the dissociation of viable cells can be problematic

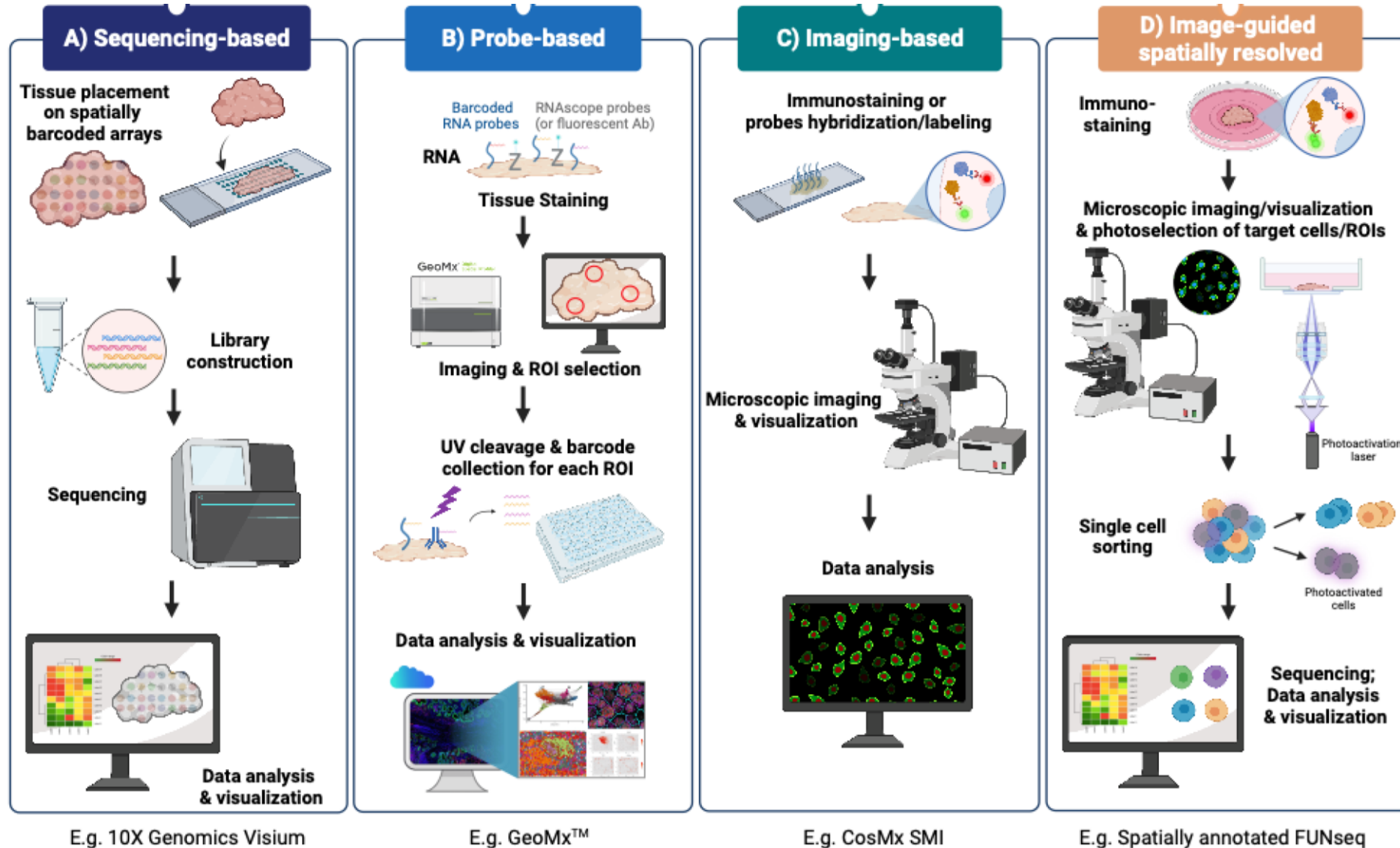
Resource

Spatial Transcriptomics and *In Situ* Sequencing to Study Alzheimer's Disease



Chen et al. 2020 *Cell*

Spatial transcriptomics covers a lot of very different methods



10x Genomics Visium Spatial Gene Expression

- Sequencing-based ST
- Spot resolution
 - 55µm diameter spots
 - 1-10 cells per spot
 - 100µm gap between spots
- Whole transcriptome
- V1
 - Poly-A capture
 - 6.5mm x 6.5mm (5000 total spots)
 - Probe-based
- V2
 - Probe-based
 - 11mm x 11mm (14000 total spots)

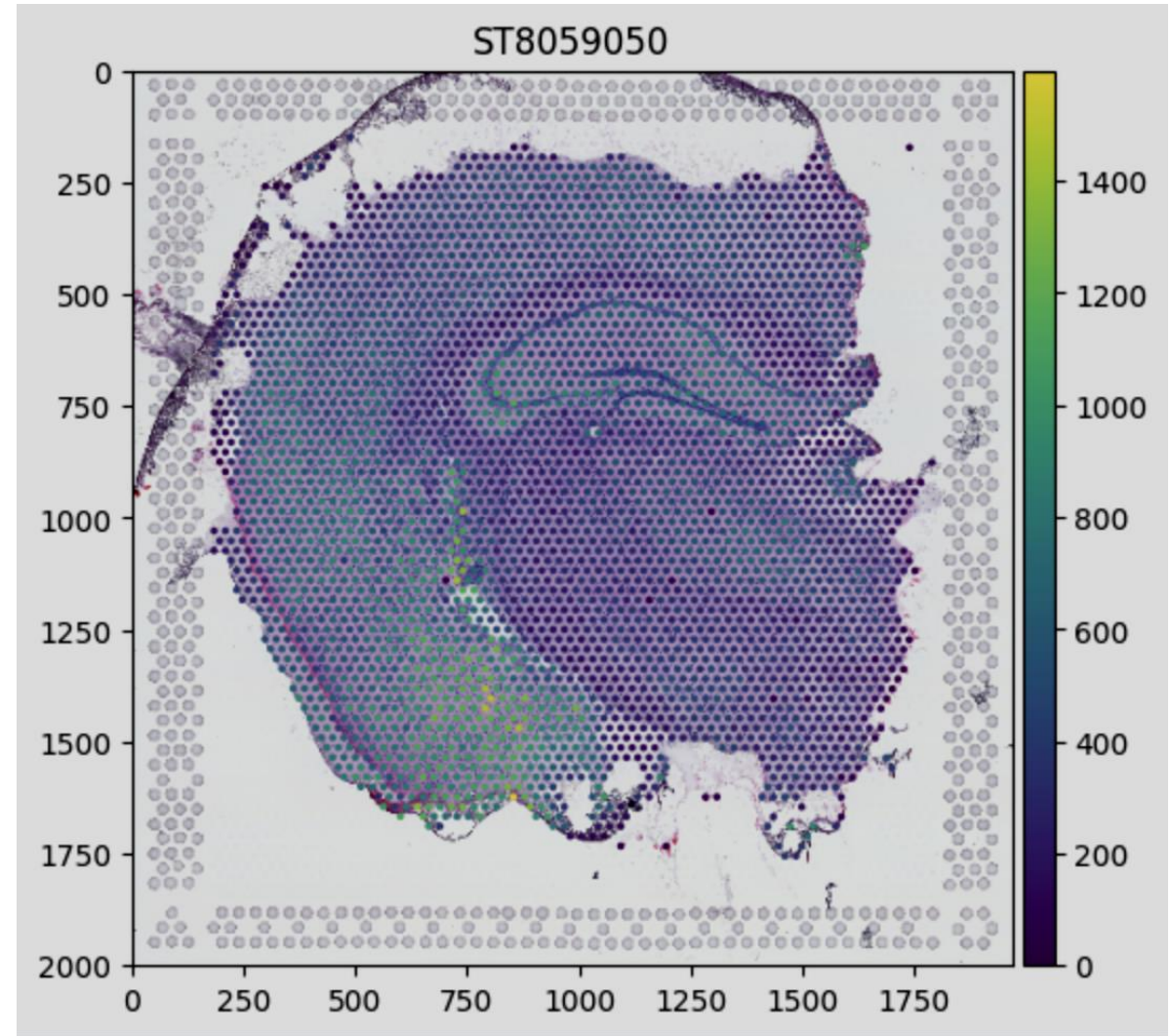
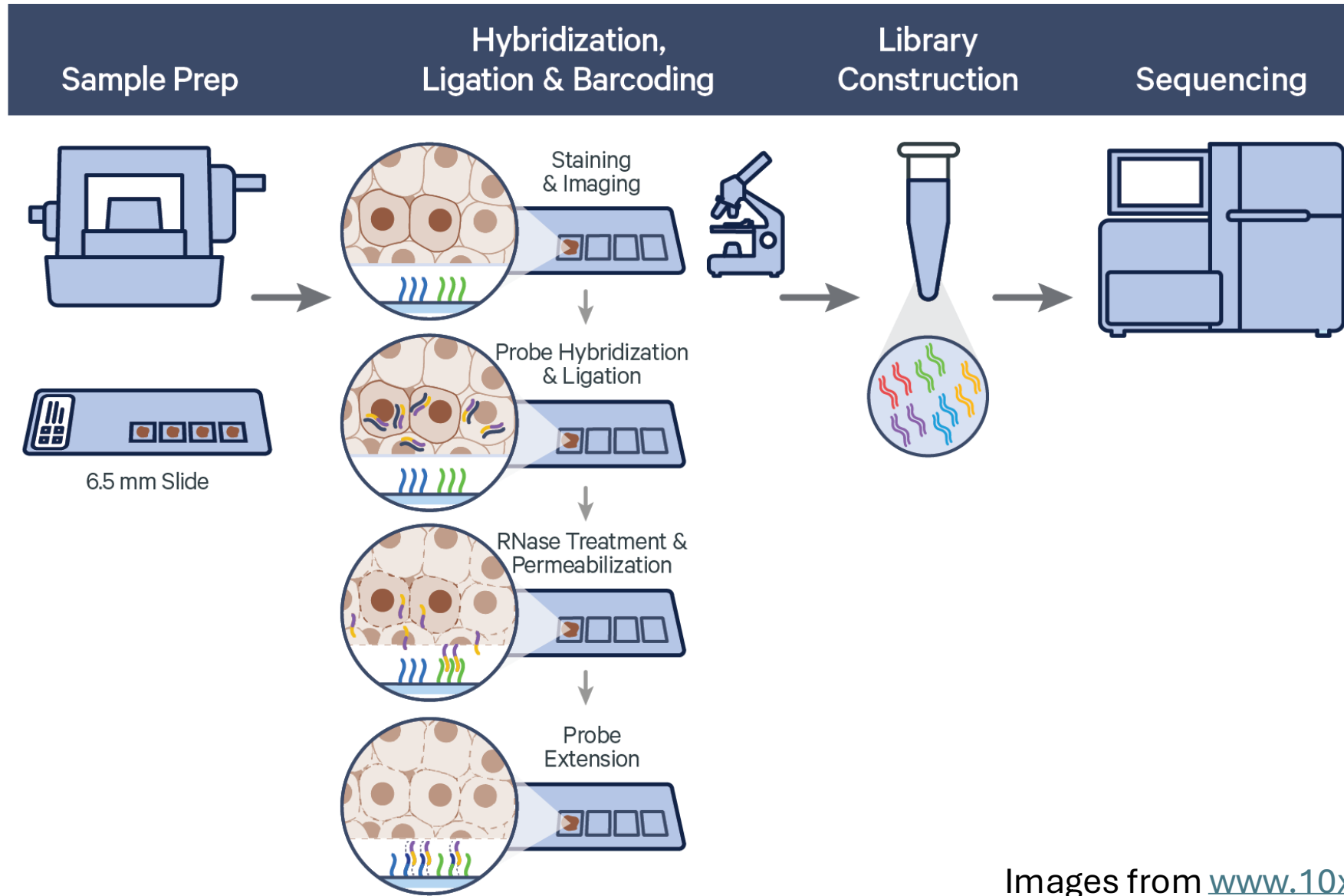


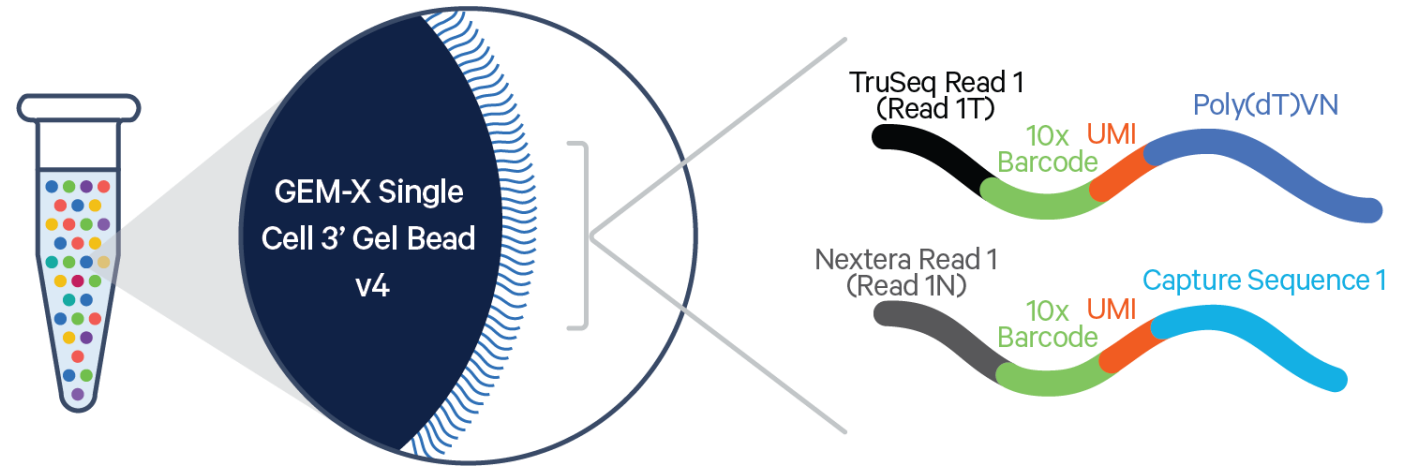
Image from: <https://spatialdata.scverse.org>

10x Genomics Visium Spatial Gene Expression

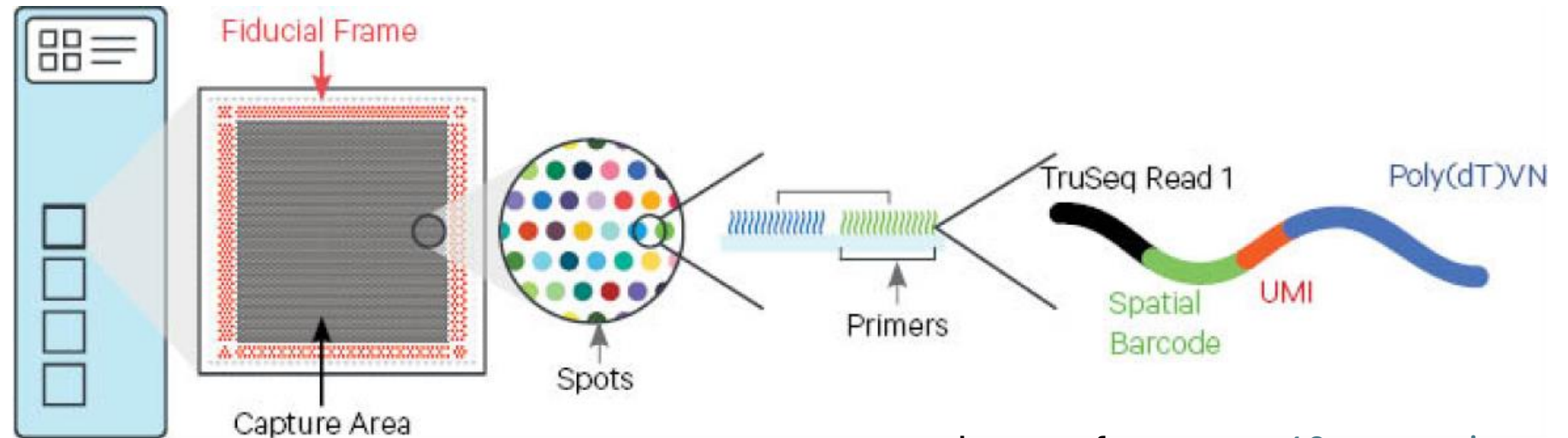


Visium reads include a barcode specific to a spot, preserving spatial information

scRNA seq

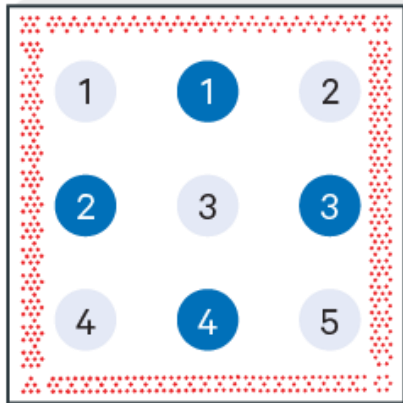


Visium

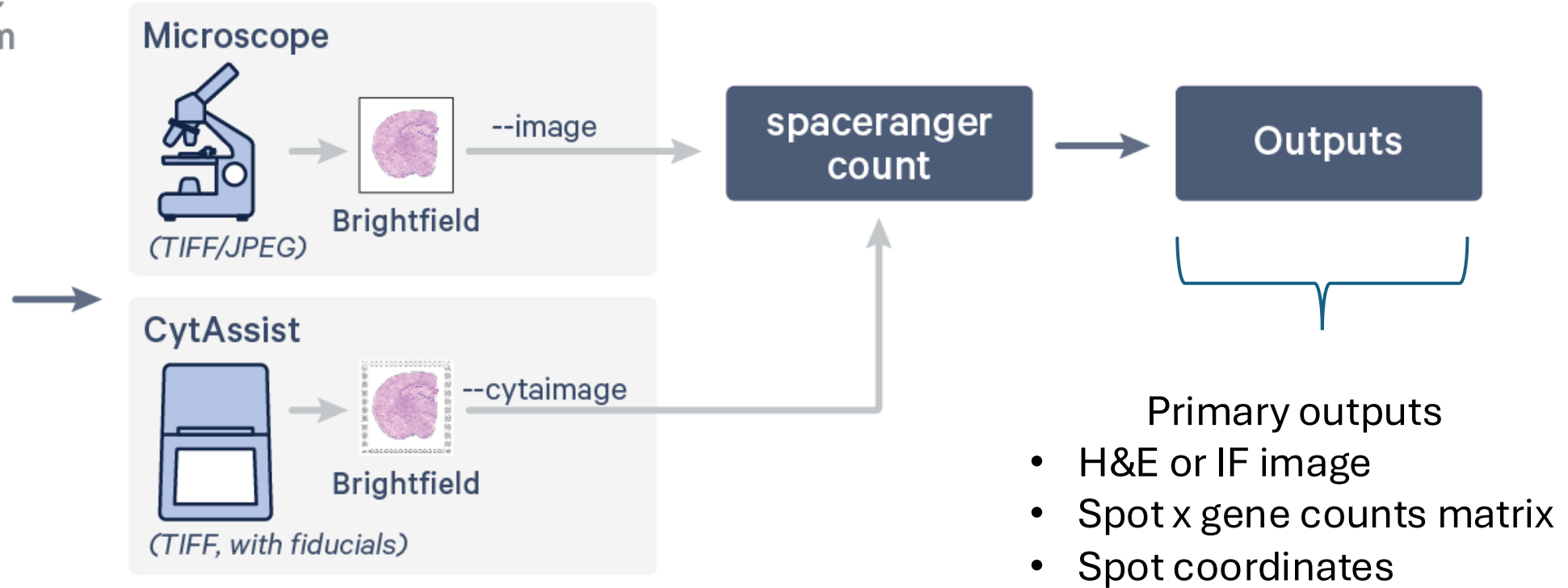


Visium datasets are processed using the SpaceRanger pipeline

CytAssist Spatial Gene Expression Slide - 11 mm

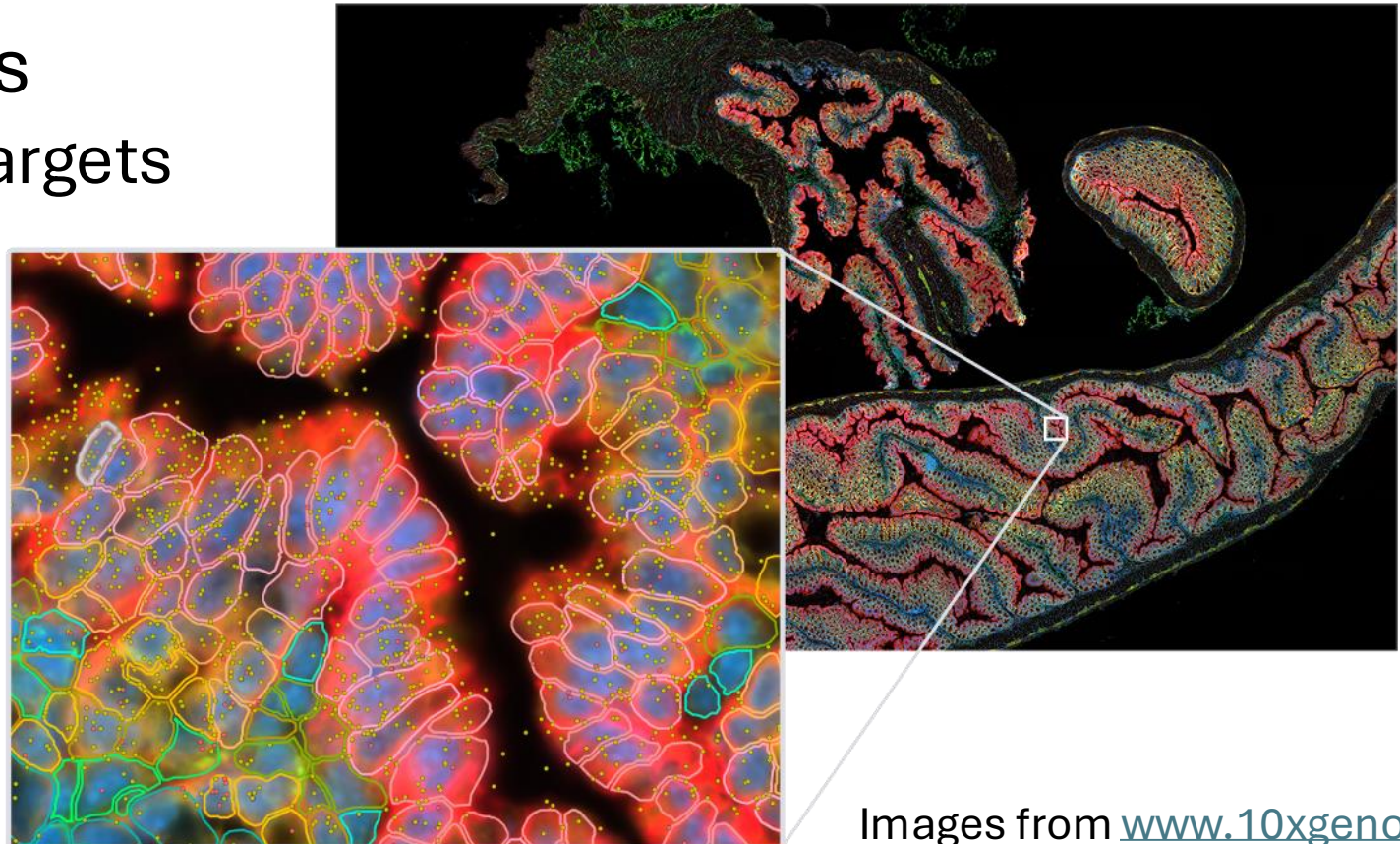


● Mouse Lung
● Mouse Brain



10x Genomics Xenium

- Imaging-based ST
- Subcellular resolution
- 12mm x 24mm capture area
- Limited gene panels
 - 500 or 5000 gene targets



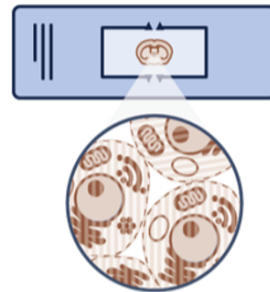
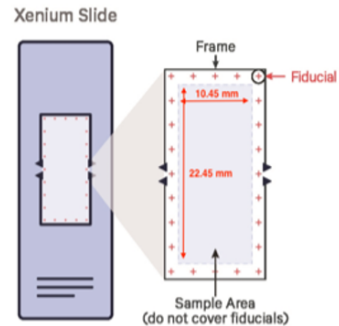
Images from www.10xgenomics.com

Xenium creates an optical barcode for each target transcript



FF or FFPE Tissue Sections
on Xenium slides

**Fixation & Permeabilization (FF) or
Deparaffinization & Decrosslinking (FFPE)**



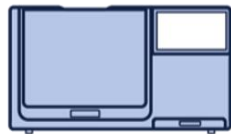
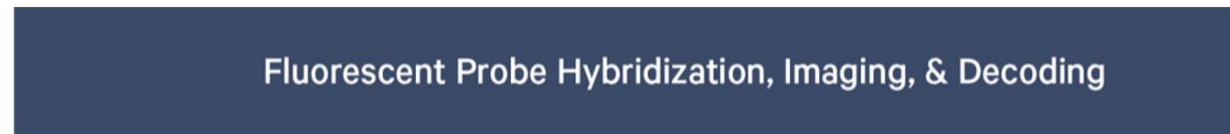
Probe Hybridization



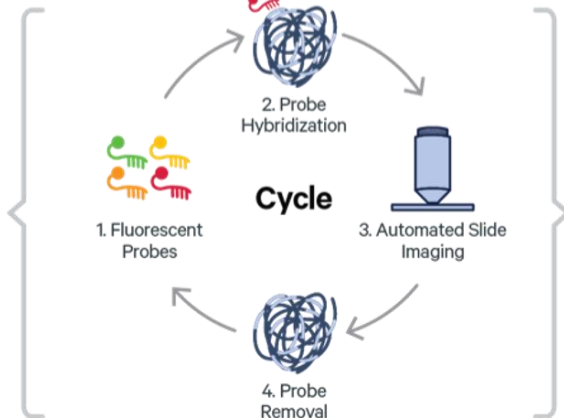
Ligation &
Primer Hybridization for Amplification



Rolling Circle Amplification
Product



**Xenium
Analyzer**

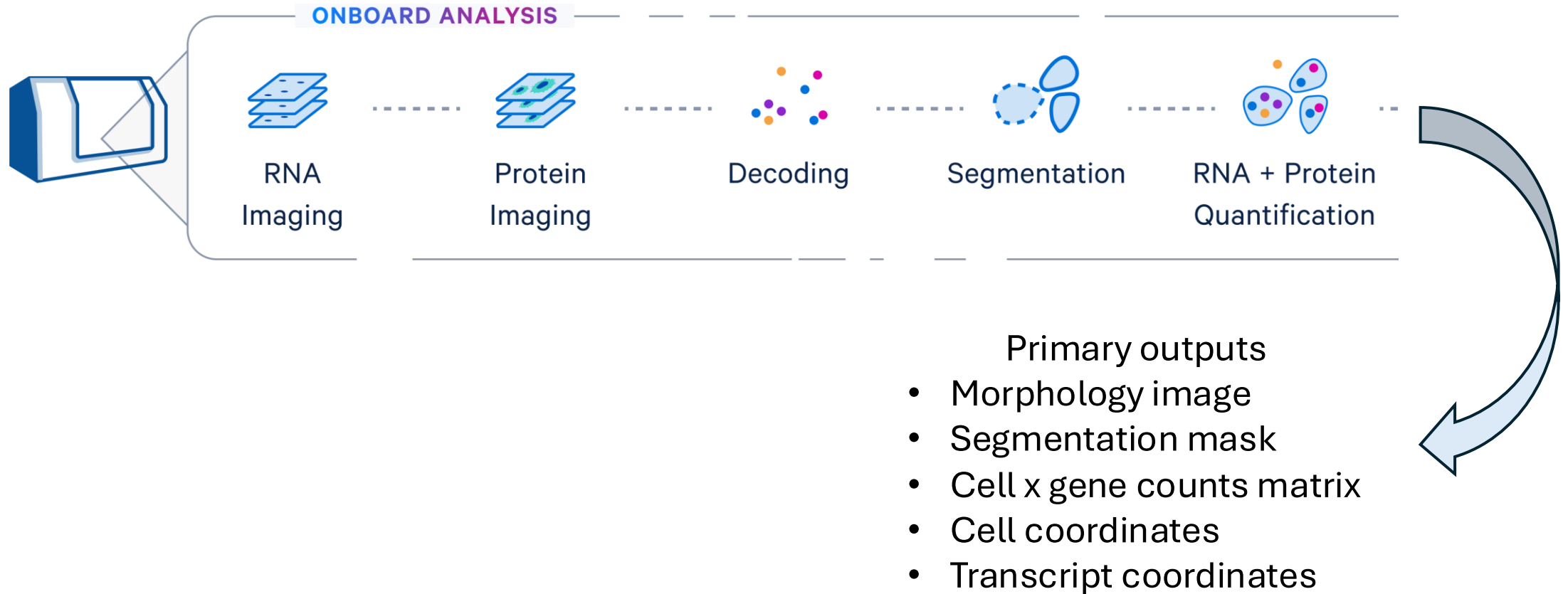


Cycle

- 1
- 2
- 3
- ⋮
- N

Images from www.10xgenomics.com

Xenium computational processing happens on the instrument and using the Xenium Ranger pipeline



QC considerations for spatial transcriptomics

- Vendors will often provide a QC report
 - [Visium web summary example](#)
 - [Xenium web summary example](#)
- QC metrics can also be calculated using Scanpy or Seurat
- Thresholds tend to be platform and tissue specific
- Standards are still being developed and benchmarked

nature biotechnology



Article

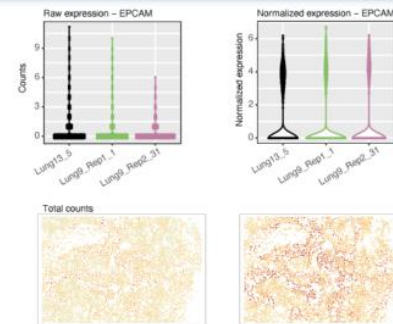
<https://doi.org/10.1038/s41587-025-02811-9>

Standardized metrics for assessment and reproducibility of imaging-based spatial transcriptomics datasets

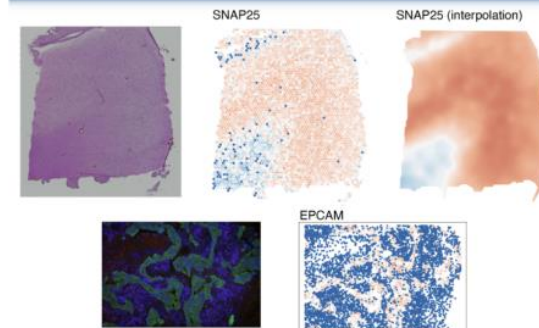
Plummer et al. (2024)

Common analysis tasks using spatial transcriptomics data

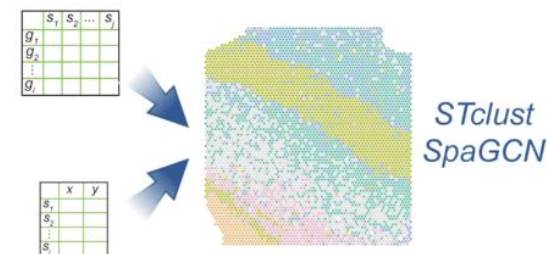
Quality control and processing



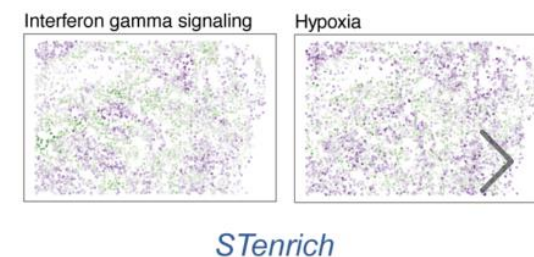
Gene expression visualization



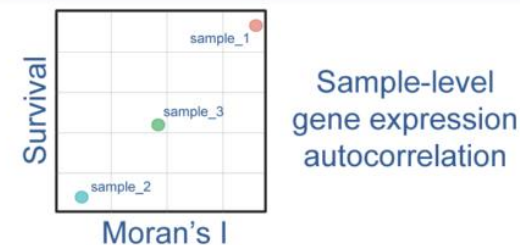
Domain detection and clustering



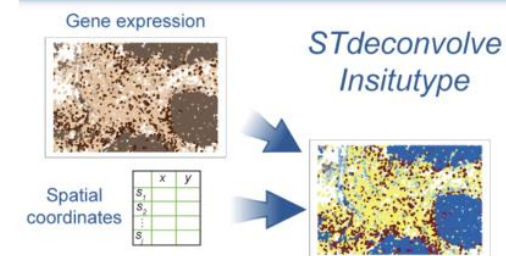
Spatial gene set enrichment



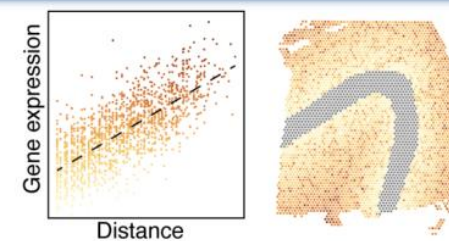
Spatial heterogeneity analysis



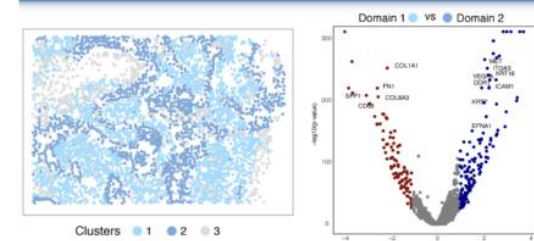
Spot/cell phenotyping



Gene expression gradients



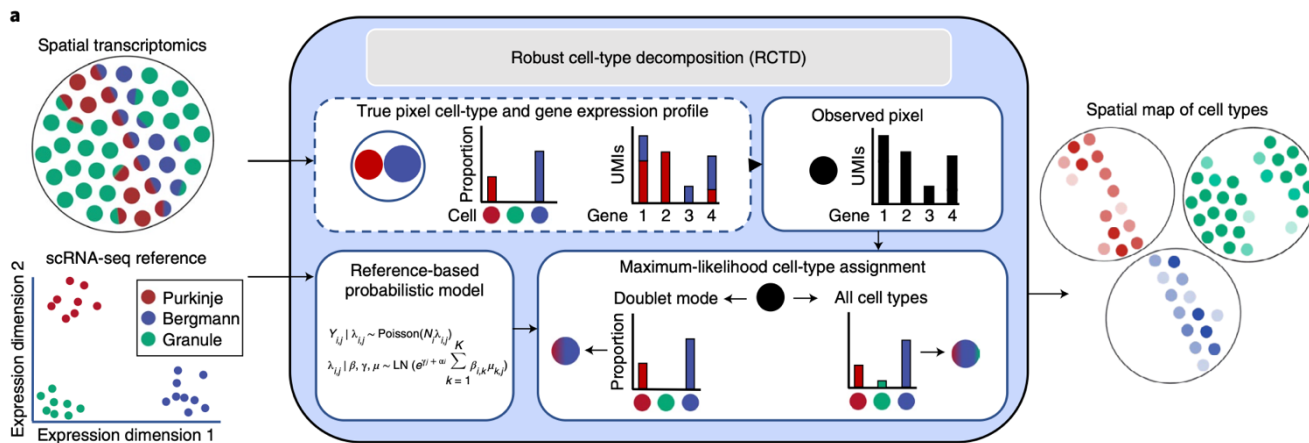
Differential gene expression



Spot deconvolution

RCTD

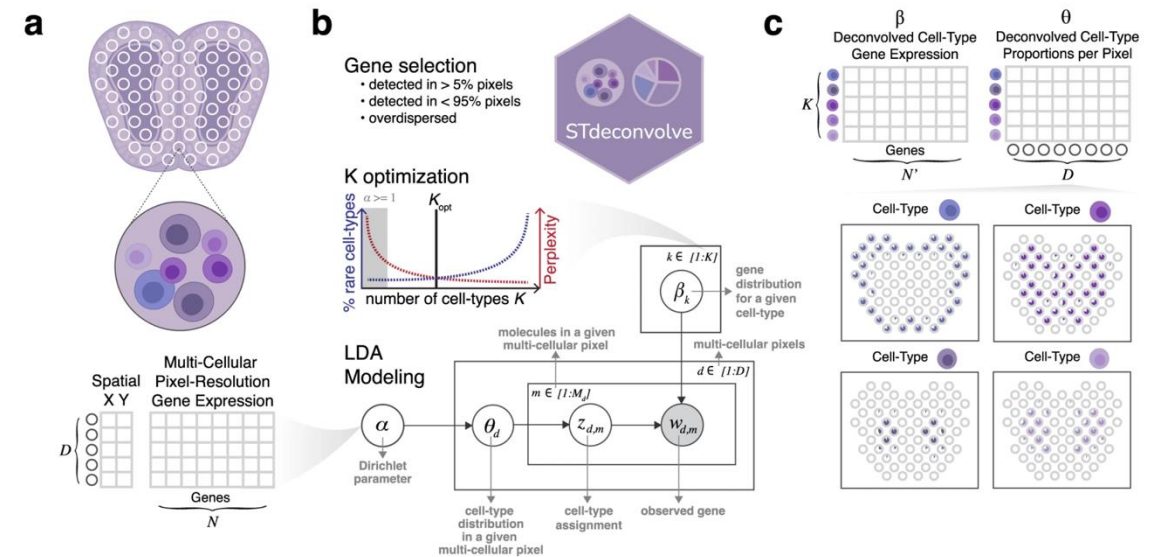
- Uses a scRNA reference atlas to define expected single-cell expression profiles and fits a statistical model to estimate the mixture of cells within a spot



Cable et al. (2022) *Nature Biotechnology*

STdeconvolve

- Uses topic modeling (LDA), developed for natural language processing, to estimate the mixture of cells within a spot



Miller et al. (2022) *Nature Communications*

High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis

